PRACTICAL – 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

1. Aim

To design a normalized MySQL database for a StudentHub application containing students, events, and registrations; create an Entity–Relationship (ER) model; implement primary and foreign key relationships; connect the database with PHP using PDO; and demonstrate secure database operations using prepared statements.

2. Problem Definition

Design the database for StudentHub with students, events and registrations. Create a PHP database connection using MySQL and implement secure database access. The database should be normalized, primary keys and foreign keys should be correctly defined, and database errors should be handled properly.

3. Key Questions / Analysis / Interpretation

Question	Interpretation
1. Is the schema normalized properly?	Yes. Student, event, and registration information is separated into different tables, reducing duplicate data and update anomalies.
2. Are primary keys and foreign keys correctly defined?	Yes. Each table has a primary key. The registrations table contains foreign keys referencing students and events.
3. Is the database connection handled securely?	Yes. PHP PDO is used with exception handling and prepared statements. User input is passed as parameters instead of being directly concatenated into SQL.

4. Conclusion

The StudentHub database was designed using a normalized relational structure containing students, events, and registrations. Primary and foreign keys were used to maintain relationships and referential integrity. The database was connected to PHP using PDO, and prepared statements were used for secure INSERT and SELECT operations. The practical demonstrates the basic concepts required for building a secure database-backed web application.

5. Output:

ID	Student ID	Name	Email	Mobile	Course	Year
1	**25DCE025**	Bhavya Patel	25dce025@charusat.edu.in	9876543210	BCA	2
2	**25DCE026**	Aarav Shah	25dce026@charusat.edu.in	9876543211	B.Tech CSE	2
3	**25DCE027**	Diya Joshi	25dce027@charusat.edu.in	9876543212	B.Tech IT	3
4	**25DCE028**	Kavya Sharma	25dce028@charusat.edu.in	9876543213	BCA	1
5	**25DCE029**	Rohan Mehta	25dce029@charusat.edu.in	9876543214	BBA	2

Table: events (3)

ID	Event Title	Date	Time	Venue	Category	Status
1	**Web Development Hackathon**	2026-11-15	09:30:00	Lab 301, FTE Building	Technical	Upcoming
2	**AI & Machine Learning Seminar**	2026-11-20	11:00:00	Auditorium A	Seminar	Upcoming
3	**Annual Tech Fest CodeCraft**	2026-12-05	10:00:00	Central Campus Ground	Cultural & Tech	Upcoming

Table: registrations (Normalized Foreign Key Join) (5)

Reg ID	Student ID	Student Name	Event Title	Registration Date	Status
1	25DCE025	Bhavya Patel	Web Development Hackathon	2026-10-20 21:58:39	Confirmed
2	25DCE025	Bhavya Patel	AI & Machine Learning Seminar	2026-10-20 21:58:39	Confirmed
3	25DCE026	Aarav Shah	Web Development Hackathon	2026-10-20 21:58:39	Confirmed
4	25DCE027	Diya Joshi	AI & Machine Learning Seminar	2026-10-20 21:58:39	Confirmed
5	25DCE028	Kavya Sharma	Annual Tech Fest CodeCraft	2026-10-20 21:58:39	Confirmed

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Light / Dark

Practical 8: Database Connection & Schema Verification

Database connection established successfully with MySQL database: studenthub_db

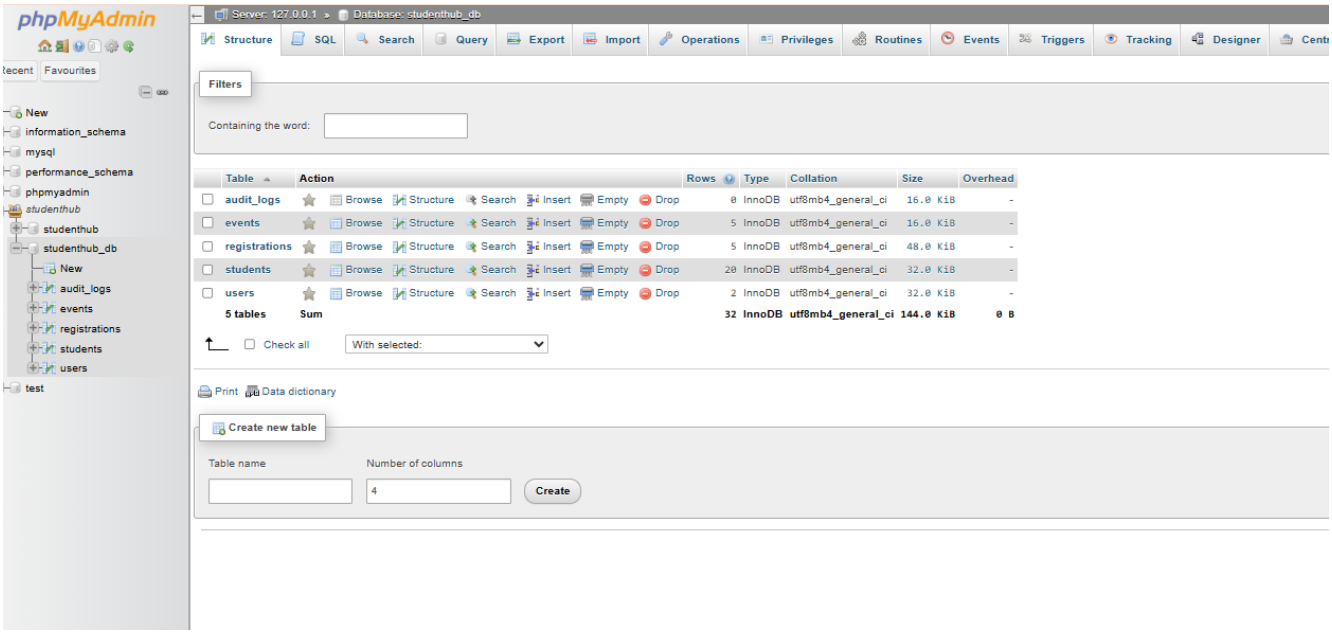
Schema Tables & Records Count

#	Table Name	Total Records	Status
1	audit_logs	0	Ready
2	events	5	Ready
3	registrations	5	Ready
4	students	20	Ready
5	users	2	Ready

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25DCE050



GitHub Repository Link

https://github.com/vatsal-1950/StudentHUB.git